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Enclosure assemblies, and related devices and methods

Abstract

A cooling plate assembly for dual-sided cooling, the cooling plate assembly can include a first cooling plate and a second cooling plate. The first cooling plate that has a first cooling plate connector side and a first cooling plate cooling side that is opposite the first cooling plate connector side. The second cooling plate that has a second cooling plate connector side and a second cooling plate cooling side that is opposite the second cooling plate connector side. The second cooling plate is configured to attach to the first cooling plate such that formed between the first cooling plate connector side and the second cooling plate connector side is at least one cooling passage through which a cooling fluid is flowed to transfer heat from both the first cooling plate cooling side and the second cooling plate cooling side to the cooling fluid.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the priority benefit of International Patent Application No. PCT/US2021/061887, Dec. 3, 2021, which claims priority to U.S. Provisional Application No. 63/120,905 filed on Dec. 3, 2020, the entire contents and disclosures of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to electrical devices for a vehicle, and more specifically, to enclosures and power conversion electrical devices for a vehicle.

BACKGROUND

(2) Propulsion and operation of electric and hybrid vehicles often involves some combination of on-board motors and batteries acting in concert. As with most vehicles, batteries used are often high-voltage, DC-power batteries, which may directly supply electrical power to certain components of the vehicle. Other components of the vehicle may require AC electrical power, and under these circumstances, a traction inverter is used to convert DC power from the batteries into AC power for these components. While often necessary in these applications, typical traction inverters can consume a fair amount of space, which along with efficiency is often a significant design constraint in any vehicle, especially electric and hybrid vehicles. Designs that reduce the space required for components such as traction inverters can provide additional space for more batteries, which can result in extended driving range and performance of electric and hybrid vehicles.

SUMMARY

(3) The present disclosure advantageously provides compact design principles, compact components, and compact arrangements of enclosure assemblies for electrical devices. More specifically, the present disclosure provides compact enclosure assemblies for traction inverters, converters, and other electrical devices used in vehicles or other applications. For example, internal electrical components of the electrical device can be compactly stacked within the enclosure assembly to thereby reduce the footprint of the electrical device. These features can be included in a robust design that is desirable in certain applications of hybrid and/or electric vehicles.

(4) In a first example, an enclosure assembly for an electrical device can include an enclosure body and a cooling plate assembly. The enclosure body can be configured to house components of the electrical device in a component cavity included in the enclosure body. The cooling plate assembly can be arranged at the component cavity and can include a first cooling plate and a second cooling plate. The first cooling plate can have a first cooling plate connector side and a first cooling plate cooling side that is opposite the first cooling plate connector side. The second cooling plate can

have a second cooling plate connector side and a second cooling plate cooling side that is opposite the second cooling plate connector side. The second cooling plate can be configured to attach to the first cooling plate such that formed between the first cooling plate connector side and the second cooling plate connector side is at least one cooling passage through which a cooling fluid is flowed to transfer heat from both the first cooling plate cooling side and the second cooling plate cooling side to the cooling fluid.

(5) In further examples of the first example, the enclosure assembly can include a flexible circuit assembly. The flexible circuit assembly can be arranged at the component cavity and attached to the cooling plate assembly. The flexible circuit assembly can include a first circuit member, a second circuit member, and a connector member that can electrically and mechanically connect the first circuit member and the second circuit member. The flexible circuit assembly can be configured to receive the cooling plate assembly between the first circuit member and the second circuit member such that the cooling plate assembly promotes cooling of both the first circuit member and the second circuit member.

(6) Continuing with the first example, a portion of the at least one cooling passage can be formed between the enclosure body and the cooling plate assembly. In examples, both a supply inlet for supplying the cooling fluid to the cooling plate assembly and a supply outlet for expelling the cooling fluid from the cooling plate assembly can be included in the enclosure body. In examples, the electrical device is a traction inverter.

(7) In a second example of the present disclosure, a cooling plate assembly for dual-sided cooling can include a first cooling plate and a second cooling plate. The first cooling plate can have a first cooling plate connector side and a first cooling plate cooling side that is opposite the first cooling plate connector side. The second cooling plate can have a second cooling plate connector side and a second cooling plate cooling side that is opposite the second cooling plate connector side. The second cooling plate can be configured to attach to the first cooling plate such that formed between the first cooling plate connector side and the second cooling plate connector side is at least one cooling passage through which a cooling fluid is flowed to transfer heat from both the first cooling plate cooling side and the second cooling plate cooling side to the cooling fluid.

(8) In further examples of the second example, the at least one cooling passage can include at least two separate cooling passages that have a shared inlet. In examples, the cooling plate assembly can be configured to receive a flow of cooling fluid at a cooling plate assembly inlet side and guide the flow of cooling fluid between the cooling plate assembly inlet side and a cooling plate assembly outlet side via the at least one cooling passage before expelling the flow of cooling fluid at the cooling plate assembly outlet side. A flow splitter can be positioned at the shared inlet and configured to split a flow of cooling fluid between the at least two separate cooling passages. In examples, each cooling passage in the at least one cooling passage can include a plurality of obstructions that is configured to influence a flow of cooling fluid through the at least one cooling passage. In examples, a first obstruction is included in an interior of the cooling passage and a second obstruction is included at a wall of the cooling passage.

(9) Continuing with the second example, at least one of the first cooling plate connector side and the second cooling plate connector side can include a groove with which to receive a sealing gasket that is configured to promote sealing of the at least one cooling passage formed between the first cooling plate connector side and the second cooling plate connector side.

(10) In some such examples, the first cooling plate can be identical to the second cooling plate. In some such examples, the at least one cooling passage can guide a flow of cooling fluid from an interior portion of the cooling plate assembly to an exterior portion of the cooling plate assembly. In examples, the at least two separate cooling passages can be symmetric about a midplane extending through and perpendicular to a length of the cooling plate assembly.

(11) In a third example of the present disclosure, a flexible circuit assembly can include a first circuit member, a second circuit member, and a connector member. The first circuit member can

have a first circuit member first side and a first circuit member second side that is opposite first circuit member first side. The second circuit member can have a second circuit member first side and a second circuit member second side. The connector member can be configured to electrically and mechanically connect the first circuit member to the second circuit member such that the first circuit member is in electrical communication with the second circuit member and so as to form a gap therebetween with which to receive a cooling plate assembly that is configured to promote cooling of both the first circuit member first side and the second circuit member first side.

(12) In further examples of the third example, at least one of the first circuit member and the second circuit member comprises a printed circuit board. In examples, the connector member is a ribbon connector. In examples, the first circuit member is the printed circuit board, and the first circuit member first side includes an insulated-gate bipolar transistor.

(13) Continuing with the third example, when the cooling plate assembly is secured between the first circuit member and the second circuit member, the first circuit member and the second circuit member can be substantially parallel to each other. In examples, a periphery of the first circuit member can include a first circuit member first edge, and a periphery of the second circuit member can include a second circuit member first edge. The first circuit member first edge and the second circuit member first edge can be coplanar when the cooling plate assembly is secured between the first circuit member and the second circuit member.

(14) Additional features and advantages of the present disclosure will become apparent to those skilled in the art upon consideration of the following detailed description of the illustrative embodiments exemplifying the disclosure as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above-mentioned and other features and advantages of this disclosure, and the manner of obtaining them, will become more apparent, and will be better understood by reference to the following description of the exemplary embodiments taken in conjunction with the accompanying drawings, wherein:

(2) FIG. 1A is a partial cutaway view of an enclosure assembly according to principles of the present disclosure;

(3) FIG. 1B is a top view of the enclosure assembly shown in FIG. 1A;

(4) FIG. 1C is a cross-sectional view of the enclosure assembly shown in FIG. 1A taken at a midsection along a height of the enclosure assembly;

(5) FIG. 1D is a cross-sectional view of the enclosure assembly shown in FIG. 1A taken at a midsection along a width of the enclosure assembly;

(6) FIG. 2A is a perspective view of a flexible circuit assembly according to principles of the present disclosure;

(7) FIG. 2B is a perspective view of the flexible circuit assembly in FIG. 2A during a folding step;

(8) FIG. 2C is a perspective view of the flexible circuit assembly in FIG. 2A after a folding step;

(9) FIG. 3A is a perspective view of a cooling plate assembly according to principles of the present disclosure;

(10) FIG. 3B is a perspective view of a first cooling plate of the cooling plate assembly shown in FIG. 3A;

(11) FIG. 3C is a perspective view of a second cooling plate of the cooling plate assembly shown in FIG. 3A;

(12) FIG. 4 is a flow diagram of cooling fluid flowing through an enclosure assembly according to principles of the present disclosure;

(13) FIG. 5 is a flowchart of a method of assembling a stack comprising both a flexible circuit

assembly and a cooling plate assembly according to principles of the present disclosure;

(14) FIG. 6A is a perspective view of an enclosure assembly with an individual printed circuit board according to principles of the present disclosure;

(15) FIG. 6B is another perspective view of the enclosure assembly shown in FIG. 6A;

(16) FIG. 6C is a cross-sectional view of the enclosure assembly shown in FIG. 6A taken at a midsection along a width of the enclosure assembly;

(17) FIG. 7A is a partial cutaway view of an enclosure assembly employing cylindrical capacitors according to principles of the present disclosure;

(18) FIG. 7B is a cross-sectional view of the enclosure assembly shown in FIG. 7A taken at a midsection along a width of the enclosure assembly;

(19) FIG. 7C is a partial cutaway view of an enclosure assembly employing box capacitors according to principles of the present disclosure;

(20) FIG. 7D is a cross-sectional view of the enclosure assembly shown in FIG. 7C taken at a midsection along a width of the enclosure assembly;

(21) FIG. 8A is a perspective view of the electrical device with the top cover removed for illustration purposes only;

(22) FIG. 8B is a perspective view of the electrical device in FIG. 8A with a flexible circuit cover removed for illustration purposes only to show a control board;

(23) FIG. 8C is a perspective view of the electrical device in FIG. 8B with the control board removed for illustration purposes only to show the first circuit member of the flexible circuit assembly;

(24) FIG. 8D is a perspective view of the electrical device in FIG. 8C from the bottom of the electrical device with the bottom cover removed for illustration purposes only to show the second circuit member in the flexible circuit board

(25) FIG. 8E is isolated perspective view of a bottom side of the cooling plate;

(26) FIG. 8F is an isolated perspective view of a top side of the cooling plate;

(27) FIG. 8G is a cross-sectional view through a midsection of the width of the electrical device shown in FIG. 8A;

(28) FIG. 8H is an isolated view of the cooling plate and body of the electrical device in FIG. 8A;

(29) FIG. 8I is a close-up view showing a leak detection channel flanked by gaskets in FIG. 8H;

and

(30) FIG. 9 is a perspective view a 6-phase electrical device.

(31) Corresponding reference characters indicate corresponding parts throughout the several views. Although the drawings represent embodiments of various features and components according to the present disclosure, the drawings are not necessarily to scale and certain features may be exaggerated in order to better illustrate and explain the present disclosure. The exemplification set out herein illustrates an embodiment of the invention, and such an exemplification is not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION OF THE DRAWINGS

(32) For the purposes of promoting an understanding of the principles of the present disclosure, reference is now made to the embodiments illustrated in the drawings, which are described below. The exemplary embodiments disclosed herein are not intended to be exhaustive or to limit the disclosure to the precise form disclosed in the following detailed description. Rather, these exemplary embodiments were chosen and described so that others skilled in the art may utilize their teachings. It is not beyond the scope of this disclosure to have a number (e.g., all) the features in a given embodiment to be used across all embodiments.

(33) FIGS. 1A-1D show various views of an electrical device, according to principles of the present disclosure. FIG. 1A shows a partial cutaway view of an enclosure assembly **100**. FIG. 1B shows a top view of the enclosure assembly **100** shown in FIG. 1A. FIG. 1C shows a cross-sectional view of the enclosure assembly **100** shown in FIG. 1A taken at a midsection along a height of the

enclosure assembly **100**. FIG. 1D shows a cross-sectional view of the enclosure assembly **100** shown in FIG. 1A taken at a midsection along a width of the enclosure assembly **100**. Although discussed in relation to one type of electrical device (e.g., a traction inverter), it should be understood that one skilled in the art will appreciate that enclosures and related enclosure methods disclosed herein are applicable across a variety of electrical devices, including power conversion (e.g., inverters, converter), computing, and other similar devices.

(34) For illustrative purposes, FIGS. 1A-1D show a traction inverter **10** that employs principles of the present disclosure. An enclosure assembly **100** for an electrical device, such as the shown traction inverter **10**, can include an enclosure body **102**, a component cavity **104** formed at the enclosure body **102**, and a cover **105** for covering the component cavity **104**. The enclosure body **102** can include a supply inlet **106** and a supply outlet **108**. The enclosure assembly **100** can include a flexible circuit assembly **110** that has a first circuit member **111** and a second circuit member **112**. As used herein, the flexible circuit assembly can refer to foldability between two semi-rigid or rigid circuit members (e.g., first and second circuit members **111**, **112**) and/or flexible circuit members (e.g., first and second circuit members **111**, **112**). As well, the enclosure assembly **100** can include a cooling plate assembly **120** that has a first cooling plate **121** and second cooling plate **122**. These components of the enclosure assembly **100** will be discussed in further detail hereinafter.

(35) The enclosure body **102** can be configured to house components of the electrical device in the component cavity **104** included in the enclosure body **102**. As shown, a flexible circuit assembly **110** connects internal electrical components **131** to each other and/or to external connectors **133**. Due to the flexibility of the flexible circuit assembly **110**, a footprint of the enclosure assembly **100** can be reduced because, for example, the flexible circuit assembly **110** can be folded within the enclosure assembly **100** (instead of having to extend longitudinally therein). The flexible circuit assembly **110** can be arranged at the component cavity **104** and, for cooling the internal electrical components **131**, attached to the cooling plate assembly **120** (the details of which will be discussed further below). The flexible circuit assembly **110** can be configured to receive the cooling plate assembly **120** between a first circuit member **111** and the second circuit member **112** such that the cooling plate assembly **120** promotes cooling of both a first circuit member **111** and the second circuit member **112**. As shown here, a capacitor **133** included in the internal electrical components **131** can be a brick-type capacitor **133**, but as discussed further hereinafter, other types of capacitors **133** may be employed based on the desires and constraints in a particular application.

(36) Temperature regulation of internal electrical components **131** ensure their proper performance. In this regard, the enclosure assembly **100** can facilitate cooling of internal electrical components **131** within the enclosure, especially those internal electrical components **131** that are most sensitive to high temperatures. The cooling plate assembly **120** can be arranged at the component cavity **104** and can include a first cooling plate **121** and a second cooling plate **122**, which mate together to form the cooling plate assembly **120**. The flexible circuit assembly **110** can attach to the cooling plate assembly **120**. For example, as discussed above, the cooling plate assembly **120** can be arranged between first and second circuit members **111**, **112** of the flexible circuit assembly **110**. In this way, a stack **140** (e.g., comprising the first circuit member **111**, the first cooling plate **121**, the second cooling plate **122**, and the second circuit member **112**) can be formed and arranged within the component cavity **104** to make with portions of the enclosure body **102**. In examples, as noted above, both the supply inlet **106** for supplying the cooling fluid to the cooling plate assembly **120** and the supply outlet **108** for expelling the cooling fluid from the cooling plate assembly **120** can be included in the enclosure body **102**.

(37) FIGS. 2A-2C show various views of a flexible circuit assembly **110** according to principles of the present disclosure. FIG. 2A shows a perspective view of the flexible circuit assembly **110**. FIG. 2B shows a perspective view of the flexible circuit assembly **110** in FIG. 2A during a folding step. FIG. 2C shows a perspective view of the flexible circuit assembly **110** in FIG. 2A after a folding

step.

(38) Flexibility of the flexible circuit assembly **110** can be encouraged by its construction. As noted above, the flexible circuit assembly **110** can be configured to receive the cooling plate assembly between the first circuit member **111** and the second circuit member **112** such that the cooling plate assembly promotes cooling of both the first circuit member **111** and the second circuit member **112**. A connector member **250** included in the flexible circuit assembly **110** can electrically and mechanically connect the first circuit member **111** and the second circuit member **112**. The connector member **250** can be flexible and can be configured to electrically and mechanically connect the first circuit member **111** to the second circuit member **112** such that the first circuit member **111** is in electrical communication with the second circuit member **112**.

(39) In an example, the connector member **250** can allow the first circuit member **111** to be folded over and/or under (as indicated by arrow “A”) the second circuit member **112** while allowing electrical communication therebetween. Under these circumstances, a gap “G” with which to receive a cooling plate assembly can be formed between the first and second circuit members **111**, **112**. The first circuit member **111** can have a first circuit member first side **214** and a first circuit member second side **215** that is opposite first second member first side, and the second circuit member **112** can have a second circuit member first side **216** and a second circuit member second side **217**. By virtue of its position between the first and second circuit members **111**, **112**, for example, the cooling plate assembly can be configured to promote cooling of both the first circuit member first side **214** and the second circuit member first side **216**.

(40) As noted above, the flexible circuit assembly **110** can include printed circuit boards **260**. Thus, in some examples, at least one of the first circuit member **111** and the second circuit member **112** can include a printed circuit board **260** (e.g., a 3- or 4-phase printed circuit board **260**). When the first circuit member **111** is the printed circuit board **260**, the first circuit member first side **214** can include an insulated-gate bipolar transistor (“IGBT”) **172**. In other examples, both the first circuit member **111** and the second circuit member **112** comprise a printed circuit board **260**. Under these circumstances, both the first circuit member first side **214** and the second circuit member first side **216** can include an IGBT **262**. In this regard, the first and second circuit members **111**, **112** can be different printed circuit boards **260** (e.g., a 4-phase IGBT board and a 3-phase IGBT board) or the same printed circuit boards **260**. Connecting the first and second circuit members **111**, **112**, for instance when both are printed circuit boards **260**, is the connector member **250** in the form of a ribbon connector **251**. Under these circumstances, the connector member **250** can attach to receptacles on both printed circuit boards **260** to thereby electrically and mechanically connect both printed circuit boards **260** for operation of the electrical device. When assembled, the connector ribbon can avoid contacting the cooling plate assembly by wrapping around the cooling plate assembly such that the cooling plate assembly and connector member **250** are spaced apart.

(41) Careful arrangement of components within the stack can ensure a compact design thereof but may also result in potential areas for thermal hotspots during operation. In examples, when the cooling plate assembly is secured between the first circuit member **111** and the second circuit member **112**, the first circuit member **111** and the second circuit member **112** can be substantially parallel to each other. In examples, a periphery of the first circuit member **111** can include a first circuit member first edge **218**, and a periphery of the second circuit member **112** can include a second circuit member first edge **219**. The first circuit member first edge **218** and the second circuit member first edge **219** can be coplanar when the cooling plate assembly is secured between the first circuit member **111** and the second circuit member **112**. This example is just one of many examples. In that regard, any edges of the flexible circuit member **111** can be aligned to fit a desired application. Arranging the first and second circuit members **111**, **112** in this fashion can result in corresponding sides of the first and second connector members that produce heat to confront one another, which can increase operating temperatures there at. During operation in this arrangement, internal electrical components such as the IGBT **262** will generate heat, which should

be removed so that the internal electrical components can operate at a safe working temperature that does not significantly detriment their performance.

(42) FIGS. 3A-3C show various views of a cooling plate assembly **120** according to principles of the present disclosure. FIG. 3A shows a perspective view of the cooling plate assembly **120**. FIG. 3B shows a perspective view of a first cooling plate **121** of the cooling plate assembly **120** shown in FIG. 3A. FIG. 3C shows a perspective view of a second cooling plate **122** of the cooling plate assembly **120** shown in FIG. 3A.

(43) Proper cooling of internal electric components can occur via the cooling plate assembly **120**. In examples, as noted above, the cooling plate assembly **120** can be arranged at the component cavity and can include a first cooling plate **121** and a second cooling plate **122**. The first cooling plate **121** can have a first cooling plate cooling side **324** and a first cooling plate connector side **325** that is opposite the first cooling plate cooling side **324**. The second cooling plate **122** can have a second cooling plate cooling side **326** and a second cooling plate connector side **327** that is opposite the second cooling plate cooling side **326**. The second cooling plate **122** can be configured to attach to the first cooling plate **121** such that formed between the first cooling plate connector side **325** and the second cooling plate connector side **327** is at least one cooling passage **370** through which a cooling fluid is flowed to transfer heat from both the first cooling plate cooling side **324** and the second cooling plate cooling side **326** to the cooling fluid. In examples, a portion of the at least one cooling passage **370** can be formed between the enclosure body and the cooling plate assembly **120** (as can be seen in FIG. 4).

(44) For manufacturing purposes, each of the first and second cooling plates **121**, **122** can be formed via casting process. For example, the first and second cooling plates **121**, **122** can be cast aluminum or another similar metal. Portions of the first and second cooling plates **121**, **122** that form the at least one cooling passage **370** can be in the form of protruding portions extending (e.g., generally upwardly) from a base **374** of at least one of the first and second cooling plates **121**, **122**. In this manner, the first and second cooling plates **121**, **122** can have mating surfaces that do not penetrate the other of the first and second cooling plates **121**, **122**. In other examples, mating surfaces of the first and second cooling plates **121**, **122** can penetrate each other in one way (e.g., the first cooling plate **121** penetrates the second cooling plate **122**) or both ways (the first and second cooling plates **121**, **122** penetrate each other). In any of these instances, walls **372** of the at least one cooling passage **370** formed between the first and second cooling plates **121**, **122** can be sealed

(45) Proper sealing of the cooling plate can prevent unintended, damaging leaks. In this regard, at least one of the first cooling plate connector side **325** and the second cooling plate connector side **327** can include a groove (not shown) with which to receive a sealing gasket (not shown) that is configured to promote sealing of the at least one cooling passage **370** formed between the first cooling plate connector side **325** and the second cooling plate connector side **327**. The sealing gasket can be any gasket made from a material now known or later invented so long as the sealing gasket seals the walls **372** of the at least one cooling passage **370**.

(46) FIG. 4 shows a flow diagram of cooling fluid flowing through an enclosure assembly **100** according to principles of the present disclosure.

(47) As noted above, cooling fluid and flow between the enclosure body **102** and the cooling plate assembly **120**. For illustration purposes, the cooling plate assembly **120** shown here has an inner portion **420a**, an outer portion **420b**, a cooling plate assembly inlet side **428**, and a cooling plate assembly outlet side **429**. The cooling plate assembly inlet side **428** can be positioned proximate to the supply inlet **106** of the enclosure body **102**, and the cooling plate assembly outlet side **429** can be positioned proximate to the supply outlet **108** of the enclosure body **102**. In examples, the cooling plate assembly **120** can be configured to receive a flow of cooling fluid at a cooling plate assembly inlet side **428** and guide the flow of cooling fluid between the cooling plate assembly inlet side **428** and a cooling plate assembly outlet side **429** via the at least one cooling passage **370**

before expelling the flow of cooling fluid at the cooling plate assembly outlet side **429**. In examples, the at least one cooling passage **370** can guide a flow of cooling fluid from an inner portion **420a** of the cooling plate assembly **120** to an outer portion **420b** of the cooling plate assembly **120**. As well, the at least one cooling passage **370** may be configured such that the rate of flow of the cooling fluid is slowed or quickened so as to vary the rate of cooling at the corresponding portion.

(48) In examples, the at least one cooling passage **370** can include at least two separate cooling passages **470a**, **470b** that have a shared inlet **472**. A flow splitter **474** can be positioned at the shared inlet **472** and configured to split a flow of cooling fluid between the at least two separate cooling passages **470a**. The flow splitter **474** can be positioned proximate to and confront the supply inlet **106** (e.g. at the cooling plate inlet side). In the illustrated embodiment, the flow splitter **474** is shown as an end of a longitudinally extending wall **372** of the at least one cooling passage **370**. It is noted that the flow splitter **474** can also be separate from walls **372** of the at least one cooling passage **370**. The thickness of the flow splitter **474** can taper toward the supply inlet **106** so as to promote splitting of the supplied flow of cooling fluid. While the illustrated embodiment shows the flow splitter **474** bisecting the fluid flow, in other embodiments the flow splitter **474** may be eccentrically located so as to vary a split of the flow of cooling fluid.

(49) In examples, each cooling passage in the at least one cooling passage **370** can include a plurality of obstructions **480** that is configured to influence a flow of cooling fluid through the at least one cooling passage **370**. Obstructions **480**, for example, can be in the form of stems **482** partially or fully extending along a height and within a cooling passage or protrusions **484** from walls **372** of the cooling pack passage. obstructions **480** may vary in size and shape depending on applications thereof obstructions **480** may be arranged in a pattern (e.g., one or more rows or one or more groupings). In examples, a first obstruction **486** is included in an interior of the at least one cooling passage **370** and a second obstruction **487** is included at a wall **372** of the at least one cooling passage **370**.

(50) Symmetry of the cooling plate assembly **120** can promote efficient manufacturing and operation processes. In examples, the first cooling plate **121** can be identical to the second cooling plate. Under these circumstances, the same cooling plate can be manufactured to be either the first cooling plate **121** or the second cooling plate. Thus, for instance, manufacturing and replacement cost can be minimized. In examples, the at least two separate cooling passages **470a** can be symmetric about a midplane extending through and perpendicular to a length of the cooling plate assembly **120**. In this manner, the cooling plate assembly **120** can promote equal distribution of the cooling fluid flow, for example when the flow splitter **474** bisects the flow of cooling fluid from the supply inlet **106**. Thus, equal cooling of the cooling plate assembly **120** can be achieved. It is noted, however, that while symmetry has a number of advantages as listed here, this disclosure also contemplates a first cooling plate **121** that is asymmetric to the second cooling plate for particular applications.

(51) FIG. 5 shows a flowchart of a method **500** of assembling a stack comprising both a flexible circuit assembly and a cooling plate assembly according to principles of the present disclosure.

(52) As alluded to above, the present disclosure also includes a method **500** of assembling a stack comprising a flexible circuit assembly and a cooling plate assembly. The flexible circuit assembly and the cooling plate assembly can be similar to those disclosed elsewhere herein, including the flexible circuit assembly **110** and the cooling plate assembly **120**. At step **502**, the method **500** can include attaching the first cooling plate to the second cooling plate. At step **504**, the method **500** can include attaching a first circuit member of the flexible circuit assembly to a first cooling plate of the cooling plate assembly. At step **506**, the method **500** can include folding the first circuit member relative to (e.g. over or under) the second circuit member. At step **508**, the method **500** can include attaching a second circuit member of the flexible circuit assembly to a second cooling plate of the cooling plate assembly. In some examples, this folding step **506** may be preceded by

attaching the first cooling plate to the second cooling plate at step **502** while in other examples, the folding step **506** may be subsequent to attaching the first cooling plate to the second cooling plate at step **502** as shown. In this manner a stack can be assembled and then arranged within an enclosure assembly. The enclosure assembly can be similar to those discussed elsewhere herein, including the enclosure assembly **100**.

(53) It is well understood that methods that include one or more steps, the order listed is not a limitation of the claim unless there are explicit or implicit statements to the contrary in the specification or claim itself. It is also well settled that the illustrated methods are just some examples of many examples disclosed, and certain steps may be added or omitted without departing from the scope of this disclosure. Such steps may include incorporating devices, systems, or methods or components thereof as well as what is well understood, routine, and conventional in the art.

(54) The remaining figures show other examples of principles of the present disclosure. FIGS. **6A-6C** show further examples of the electrical device **10** according to principles of the present disclosure. FIG. **6A** shows a perspective view of an enclosure assembly **100** with an individual printed circuit board **260**. FIG. **6B** shows another perspective view of the enclosure assembly **100** shown in FIG. **6A**. FIG. **6C** shows a cross-sectional view of the enclosure assembly **100** shown in FIG. **6A** taken at a midsection along a width of the enclosure assembly **100**.

(55) For example, FIGS. **6A-6C** show an embodiment of the enclosure assembly **100** where an individual printed circuit board **260** is employed in the stack **140** instead of a flexible circuit board. With this example, the individual printed circuit board **260** can be attached to one side (e.g., the first cooling plate cooling side **324**) of the cooling plate assembly **120** to form the stack **140**, and the stack **140** can be arranged within the enclosure cavity. As shown, the cooling plate assembly **120** is attached to the cover **105**, but it is noted that the cooling plate assembly **120** can be attached elsewhere in the enclosure assembly **100**.

(56) In addition, FIGS. **7A-7D** show examples of the electrical device **10** with varied capacitors **133** according to principles of the present disclosure. FIG. **7A** shows a partial cutaway view of an enclosure assembly **100** employing cylindrical capacitors **133**. FIG. **7B** shows a cross-sectional view of the enclosure assembly **100** shown in FIG. **7A** taken at a midsection along a width of the enclosure assembly **100**. FIG. **7C** shows a partial cutaway view of an enclosure assembly **100** employing box capacitors **133** according to principles of the present disclosure. FIG. **7D** shows a cross-sectional view of the enclosure assembly **100** shown in FIG. **7C** taken at a midsection along a width of the enclosure assembly **100**.

(57) FIGS. **7A-7H** show variations of capacitors **133** that are included in the internal electrical components **131** of the electrical device **10**. The electrical device **10** shown here can be similar to those electrical devices discussed elsewhere herein. As previously discussed in relation to in FIGS. **1A-1D**, the capacitor **133** can be a brick-type capacitor **133**, but may be other types of capacitors **133** in different applications. For example, as shown in FIGS. **7A-7B**, the capacitor **133** can be a cylindrical capacitor **133**. As shown in FIGS. **7C-7D**, the capacitor **133** can be a box capacitor **133**. Depending on constraints (e.g., physical and electrical constraints), desires, and operational demand in a given application, for example, one or more of these capacitor **133** examples may be preferable.

(58) Variations to the arrangement of capacitors within the electrical device **10** can be seen in FIGS. **8A-8D**. The capacitors **133** in FIGS. **7A-7D** are arranged to project inwardly (e.g., attached to a rear wall) into the electrical device **10** whereas the capacitors **133** shown in FIGS. **8A-8D** are arranged to project upwardly (e.g., attached to a floor or other surface generally parallel with the floor). In examples, a height of the capacitors **133** can be greater than a width of the capacitors **133**. In this regard, the additional footprint reduction can be achieved by arranging the capacitors **133** as shown in FIGS. **8A-8D**.

(59) Other variations of the electrical device **10** can be seen between FIGS. **8A-8I**. In particular,

FIG. 8A is a perspective view of the electrical device **10** with the top cover removed for illustration purposes only. FIG. 8B is a perspective view of the electrical device **10** in FIG. 8A with a flexible circuit cover removed for illustration purposes only to show a control board **810**. FIG. 8C is a perspective view of the electrical device **10** in FIG. 8B with the control board removed for illustration purposes only to show the first circuit member **111** of the flexible circuit assembly **110**. FIG. 8D is a perspective view of the electrical device **10** in FIG. 8C from the bottom of the electrical device **10** with the bottom cover removed for illustration purposes only to show the second circuit member **112** in the flexible circuit board. FIG. 8E is isolated perspective view of a bottom side of the cooling plate. FIG. 8F is an isolated perspective view of a top side of the cooling plate. FIG. 8G is a cross-sectional view through a midsection of the width of the electrical device **10** shown in FIG. 8A. FIG. 8H is an isolated view of the cooling plate and body of the electrical device **10** in FIG. 8A. FIG. 8I is a close-up view showing a leak detection channel flanked by gaskets in FIG. 8H.

(60) As can be seen in these figures, the first and second circuit members **111**, **112** can be arranged in a stacked configuration similar to those disclosed elsewhere herein. These circuit members, however, can be in electrical communication by connecting to the control board **810**. In this regard, the control board **810** can be common to the first and second circuit members **111**, **112** and operate similarly to ribbon connectors described elsewhere herein. In addition, the control board **810** can facilitate governance of the first and second members, e.g., by relaying control signals from one or more processors (not shown) connected to the control board **810**. This arrangement of the first and second circuit members **111**, **112** is just one example of the many examples disclosed herein as one skilled in the art will appreciate.

(61) As best seen in FIGS. 8E-8F, the cooling plate assembly **120** can include one or more bosses with which to facilitate mounting of connected components. For instance, the control board and first and second circuit members **111**, **112** can connect to the cooling plate via the one or more bosses. Of course, the degree to which these one or more bosses protrude outwardly from the cooling plate assembly **120** can vary based on desired positioning of the connected components.

(62) As best seen in FIGS. 8G-8I, the cooling plate assembly **120** can include a double seal **910** for cooling plate leak detection. In general, a leak detection channel **912** can be disposed between the seals **910** in the double seal **910**. These components can abut portions of the body or other internal components of the electrical device **10** and be circumferentially arranged about the periphery of the cooling plate assembly **120**, outside of the cooling passages **370**. In this regard, a leak can be detected if the seal **910** (e.g., a gasket or other resilient seal) closest to the cooling passages **370** fails via the leak detection channel **912**. Further, the seal furthest from the cooling passages **370** can maintain a seal to provide leak protection.

(63) FIG. 9 shows a 6-phase electrical device **10** rather than a 7-phase electrical device **10** as illustrated elsewhere herein. This electrical device **10** can otherwise be similar in all respects to those disclosed elsewhere herein. As one skilled in the art will appreciate, the number of phases **915** included in the electrical device **10** can vary from one to a plurality (e.g., 2, 5, 6, 7, 8 10, etc.) of phases without departing from the scope of this disclosure.

(64) The connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements. The scope is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an

embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B or C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C.

(65) In the detailed description herein, references to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art with the benefit of the present disclosure to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(66) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f), unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus

(67) While the present disclosure has been described as having an exemplary design, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practices in the art to which this invention pertains.

Claims

1. An assembly, comprising: a cooling plate assembly for dual-sided cooling, the cooling plate assembly comprising: a first cooling plate that has a first cooling plate connector side and a first cooling plate cooling side that is opposite the first cooling plate connector side; and a second cooling plate that has a second cooling plate connector side and a second cooling plate cooling side that is opposite the second cooling plate connector side, the second cooling plate configured to attach to the first cooling plate such that formed between the first cooling plate connector side and the second cooling plate connector side is at least one cooling passage through which a cooling fluid is flowed to transfer heat from both the first cooling plate cooling side and the second cooling plate cooling side to the cooling fluid; and a flexible circuit assembly, comprising: a first circuit member that has a first circuit member first side and a first circuit member second side that is opposite the first circuit member first side; a second circuit member that has a second circuit member first side and a second circuit member second side; and a connector member that is configured to electrically and mechanically connect the first circuit member to the second circuit member such that the first circuit member is in electrical communication with the second circuit member and so as to form a gap therebetween with which to receive the cooling plate assembly that is configured to promote cooling of both the first circuit member first side and the second circuit member first side, wherein at least one of the first cooling plate connector side and the second cooling plate connector side includes a groove with which to receive a sealing gasket that is configured to promote sealing of the at least one cooling passage formed between the first cooling plate connector side and the second cooling plate connector side.

2. The flexible circuit assembly of claim 1, wherein at least one of the first circuit member and the

second circuit member comprises a printed circuit board.

3. The flexible circuit assembly of claim 2, wherein the first circuit member is the printed circuit board, and the first circuit member first side includes an insulated-gate bipolar transistor.

4. The flexible circuit assembly of claim 1, when the cooling plate assembly is secured between the first circuit member and the second circuit member, the first circuit member and the second circuit member are substantially parallel to each other.

5. The flexible circuit assembly of claim 1, wherein the connector member is a ribbon connector.

6. The flexible circuit assembly of claim 1, wherein a periphery of the first circuit member includes a first circuit member first edge, wherein a periphery of the second circuit member includes a second circuit member first edge, and wherein the first circuit member first edge and the second circuit member first edge are coplanar when the cooling plate assembly is secured between the first circuit member and the second circuit member.

7. The cooling plate assembly of claim 1, wherein the at least one cooling passage comprises at least two separate cooling passages that have a shared inlet, and wherein a flow splitter is positioned at the shared inlet and configured to split a flow of the cooling fluid between the at least two separate cooling passages.

8. The cooling plate assembly of claim 7, wherein the at least two separate cooling passages are symmetric about a midplane extending through and perpendicular to a length of the cooling plate assembly.

9. The cooling plate assembly of claim 1, wherein each cooling passage in the at least one cooling passage includes a plurality of obstructions that is configured to influence a flow of the cooling fluid through the at least one cooling passage.

10. The cooling plate assembly of claim 9, wherein a first obstruction is included in an interior of the at least one cooling passage and a second obstruction is included at a wall of the at least one cooling passage.

11. The cooling plate assembly of claim 1, wherein the first cooling plate is identical to the second cooling plate.

12. The cooling plate assembly of claim 1, wherein the at least one cooling passage guides a flow of the cooling fluid from an inner portion of the cooling plate assembly to an outer portion of the cooling plate assembly.

13. The cooling plate assembly of claim 1, wherein the cooling plate assembly is configured to receive a flow of the cooling fluid at a cooling plate assembly inlet side and guide the flow of the cooling fluid between the cooling plate assembly inlet side and a cooling plate assembly outlet side via the at least one cooling passage before expelling the flow of the cooling fluid at the cooling plate assembly outlet side.

14. An enclosure assembly for an electrical device, the enclosure assembly comprising: an enclosure body that is configured to house components of the electrical device in a component cavity included in the enclosure body; and the assembly of claim 1, a cooling plate assembly for dual-sided cooling, the cooling plate assembly arranged at the component cavity and comprising: a first cooling plate that has a first cooling plate connector side and a first cooling plate cooling side that is opposite the first cooling plate connector side; and a second cooling plate that has a second cooling plate connector side and a second cooling plate cooling side that is opposite the second cooling plate connector side, the second cooling plate configured to attach to the first cooling plate such that formed between the first cooling plate connector side and the second cooling plate connector side is at least one cooling passage through which a cooling fluid is flowed to transfer heat from both the first cooling plate cooling side and the second cooling plate cooling side to the cooling fluid.

15. The enclosure assembly of claim 14, wherein a portion of the at least one cooling passage is formed between the enclosure body and the cooling plate assembly.

16. The enclosure assembly of claim 14, wherein both a supply inlet for supplying the cooling fluid

to the cooling plate assembly and a supply outlet for expelling the cooling fluid from the cooling plate assembly are included in the enclosure body.

17. The enclosure assembly of claim 14, wherein the electrical device is a traction inverter.
